

Ambulance Dispatch Application

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Ambulance Dispatch Design

Overall Operation

I am building two Python ambulance dispatch prototypes. Both currently load the simulation data and preview the first 10 calls in dispatch order. I have added the dispatch structure, but I still need to choose and implement the two routing algorithms. Once routing is ready, each prototype will compare routes from every ambulance's staging location, include traffic delays, and select the ambulance with the lowest travel time. It will append the call ID, type, location, selected ambulance, route, and travel time to `/var/log/ambulance_call_log.csv`, then reset the ambulance to its staging location. A timer will measure the total time spent calculating routes.

Simulation Files

Both files use Python's csv module to read the four CSVs from the data folder. The `ambulance.csv` file supplies ambulance numbers and staging locations, `call_priority.csv` supplies priorities by call type, and `calls.csv` supplies call IDs, locations, and types. The `location_network.csv` file becomes a directed graph that stores distance and travel time plus traffic delay for each connection. The code checks required fields, numeric values, duplicate IDs, call types, and locations before processing calls.

Call Priorities

Both prototypes process priority 1 calls first, followed by priority 2 and then priority 3. The code looks up each call's priority by its call type and sorts calls by priority and original row position. Since `calls.csv` has no arrival timestamps, I use file order as the order received. This keeps calls with the same priority in arrival order and gives both prototypes the same sequence for comparison.